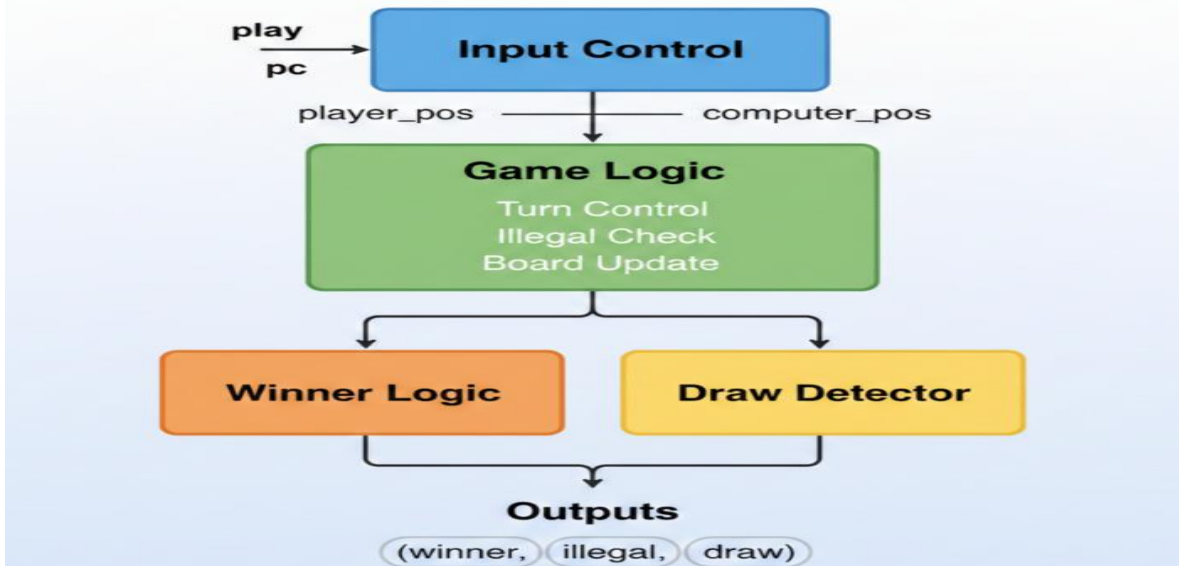




tic_tac_toe - [C:/Users/Gowthami/Documents/PROJECTS/tic_tac_toe/tic_tac_toe.xpr] - Vivado 2025.1.1

File Edit Flow Tools Reports Window Layout View Help Q: Quick Access Synthesis and Implementation Out-of-date details

Flow Navigator PROJECT MANAGER - tic_tac_toe

Project Summary x tic_tac_toe.game.v *

C:/Users/Gowthami/Documents/PROJECTS/tic_tac_toe/tic_tac_toe.srccs/sources_1/new/tic_tac_toe.game.v

```
1 module tic_tac_toe_game_v
2   input clk,
3   input reset,
4   input play,
5   input pe,
6   input [3:0] player_pos,
7   input [3:0] computer_pos,
8   output reg [1:0] winner,
9   output reg illegal,
10  output reg draw
11;
12 reg [1:0] board [0:0];
13 reg turn;
14 integer i;
15 always @(posedge clk or posedge reset)
16 begin
17   if(reset)
18   begin
19     for(i=0;i<9;i=i+1)
20       board[i] <= 2'b00;
21     winner <= 2'b00;
22     illegal <= 1'b0;
23     draw <= 1'b0;
24     turn <= 1'b0;
25   end
26   else
27   begin
28     illegal <= 1'b0;
29     if(play && turn==0)
30     begin
31       if(player_pos==3 && player_pos==5)
32         board
33       else
34         board
35     end
36   end
37 end
```

tic_tac_toe - [C:/Users/Gowthami/Documents/PROJECTS/tic_tac_toe/tic_tac_toe.xpr] - Vivado 2025.1.1

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Flow Navigator PROJECT MANAGER - tic_tac_toe

Project Summary x tic_tac_toe.game.v *

C:/Users/Gowthami/Documents/PROJECTS/tic_tac_toe/tic_tac_toe.srccs/sources_1/new/tic_tac_toe.game.v

```
32 end
33 always @(*)
34 begin
35   winner = 2'b00;
36   if(board[0]!=0 && board[0]==board[1] && board[1]==board[2])
37     winner = board[0];
38   else if(board[3]!=0 && board[3]==board[4] && board[4]==board[5])
39     winner = board[3];
40   else if(board[6]!=0 && board[6]==board[7] && board[7]==board[8])
41     winner = board[6];
42   else if(board[0]!=0 && board[0]==board[3] && board[3]==board[6])
43     winner = board[0];
44   else if(board[1]!=0 && board[1]==board[4] && board[4]==board[7])
45     winner = board[1];
46   else if(board[2]!=0 && board[2]==board[5] && board[5]==board[8])
47     winner = board[2];
48   else if(board[0]!=0 && board[0]==board[4] && board[4]==board[8])
49     winner = board[0];
50   else if(board[2]!=0 && board[2]==board[4] && board[4]==board[6])
51     winner = board[2];
52 end
53 always @(*) begin
54   if(board[0]!=0 && board[1]!=0 && board[2]!=0 &&
55     board[3]!=0 && board[4]!=0 && board[5]!=0 &&
56     board[6]!=0 && board[7]!=0 && board[8]!=0 &&
57     winner==0)
58     draw = 1'b1;
59   else
60     draw = 1'b0;
61 end
62 endmodule
```

